BrightCart

E-COMMERCE RETAILER

ANALYSIS REPORT

E-Commerce Profitability Analysis

Analyzing profitability across products, sales channels, and marketing performance to support data-driven business decisions



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PROBLEM STATEMENT

Problem

BrightCart has generated over $1M in gross revenue over the past two years, but declining net margins raise concerns about whether all product categories and sales channels are truly profitable after accounting for product costs, shipping, returns, platform fees, and marketing spend.

Highlight / Key Challenge

Understanding where the company is actually making or losing money requires analyzing profitability across product categories and sales channels, while also evaluating return rates and marketing performance.

Analysis Objective

This analysis aims to uncover actionable insights from order-level, product cost, and marketing data to identify profitable and loss-making segments, evaluate marketing effectiveness, and support data-driven decisions to improve overall profitability.

WORKFLOW OF ANALYSIS

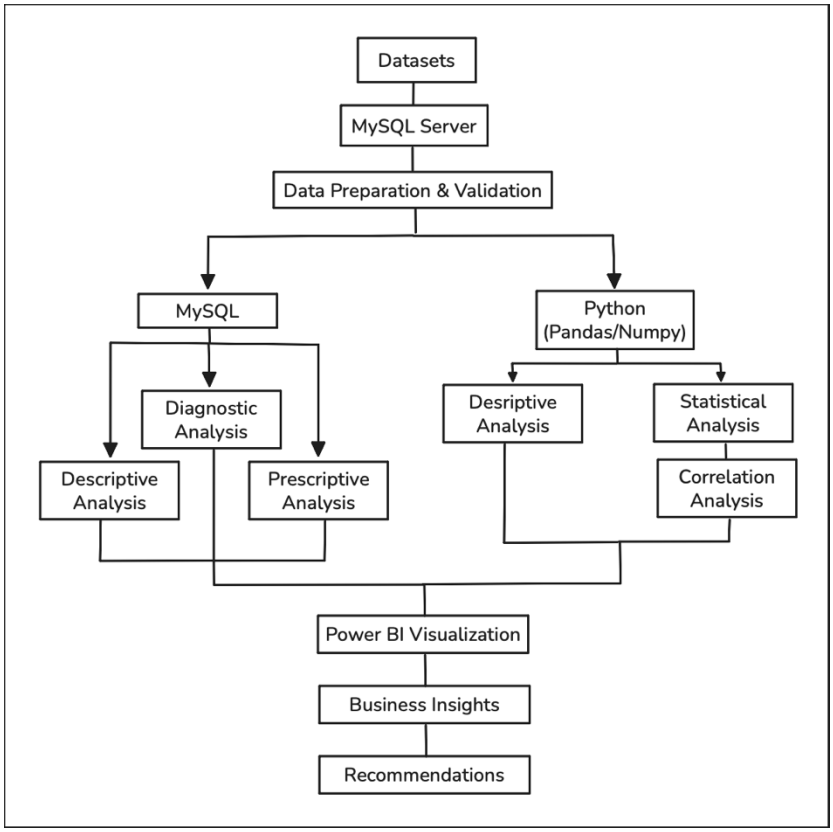


Figure 1. Workflow of Analysis

DATASET OVERVIEW

product_id	product_name	category	sub_category	unit_cost	selling_price	shipping_cost_per_unit	weight_lbs	supplier
SKU-0001	Headphones - Variant 1	Electronics	Headphones	29.44	56.08	13.42	12.4	Supplier A
SKU-0002	Headphones - Variant 2	Electronics	Headphones	26.92	91.57	15.5	15.0	Supplier B
SKU-0003	Headphones - Variant 3	Electronics	Headphones	15.91	43.34	9.58	7.6	Supplier C
SKU-0004	Headphones - Variant 4	Electronics	Headphones	7.13	24.31	5.74	2.8	Supplier B
SKU-0005	Headphones - Variant 5	Electronics	Headphones	43.05	135.82	5.82	2.9	Supplier C
SKU-0006	Phone Cases - Variant 1	Electronics	Phone Cases	11.54	35.03	9.98	8.1	Supplier B
SKU-0007	Phone Cases - Variant 2	Electronics	Phone Cases	21.92	34.0	13.66	12.7	Supplier E
SKU-0008	Phone Cases - Variant 3	Electronics	Phone Cases	72.59	192.38	10.06	8.2	Supplier A
SKU-0009	Chargers - Variant 1	Electronics	Chargers	60.74	193.42	4.78	1.6	Supplier D
SKU-0010	Chargers - Variant 2	Electronics	Chargers	67.05	222.97	12.94	11.8	Supplier A
SKU-0011	Chargers - Variant 3	Electronics	Chargers	25.39	68.98	14.54	13.8	Supplier A
SKU-0012	Chargers - Variant 4	Electronics	Chargers	31.36	78.79	13.5	12.5	Supplier D
SKU-0013	Smart Watches - Variant 1	Electronics	Smart Watches	21.44	51.28	12.38	11.1	Supplier D
SKU-0014	Smart Watches - Variant 2	Electronics	Smart Watches	5.96	20.67	4.3	1.0	Supplier A
SKU-0015	Smart Watches - Variant 3	Electronics	Smart Watches	60.94	182.6	8.78	6.6	Supplier C
SKU-0016	Smart Watches - Variant 4	Electronics	Smart Watches	78.92	260.5	11.82	10.4	Supplier A
SKU-0017	Speakers - Variant 1	Electronics	Speakers	66.69	185.03	5.42	2.4	Supplier E
SKU-0018	Speakers - Variant 2	Electronics	Speakers	11.89	37.71	14.46	13.7	Supplier D
SKU-0019	Speakers - Variant 3	Electronics	Speakers	48.17	73.73	9.34	7.3	Supplier A

Figure 2. Dataset products.csv

Dataset information:

1. The dataset products.csv consist of 207 rows
2. With the product_id has unique ID for each row
3. Has relationship/matching column “primary_category”

order_id	customer_id	order_date	channel	payment_me	region	items_ordere	primary_cate	gross_revenu	discount_pct	discount_am	shipping_cos	product_cost	platform_fee	transaction_	returned	refund_amou	net_revenue	total_costs	profit
ORD-000001	C-0617	6/24/25	Mobile App	Gift Card	Northeast	2	Electronics	261.39	0	0	22.28	92.3	0	7.88	No	0	261.39	122.46	138.93
ORD-000002	C-0720	8/23/25	Website	Credit Card	Midwest	1	Beauty	24.02	0	0	3.98	10.75	0	1	No	0	24.02	15.73	8.29
ORD-000003	C-0189	7/20/25	Website	Credit Card	Southeast	2	Toys	52.6	0	0	14.6	31.41	0	1.83	No	0	52.6	47.84	4.76
ORD-000004	C-0058	7/17/25	Mobile App	Debit Card	Southeast	2	Sports	30.63	5	1.53	30.98	16.27	0	1.14	No	0	29.1	48.39	-19.29
ORD-000005	C-0580	10/11/25	Marketplace	Credit Card	Northeast	1	Sports	11.99	25	3	10.3	6.88	1.35	0.56	No	0	8.99	19.09	-10.1
ORD-000006	C-0678	9/24/24	Mobile App	Debit Card	Northeast	4	Toys	126.99	15	19.05	61.56	57.74	0	3.43	No	0	107.94	122.73	-14.79
ORD-000007	C-0484	3/30/25	Marketplace	Credit Card	Southeast	2	Clothing	298.36	0	0	20.84	105.76	44.75	8.95	No	0	298.36	180.3	118.06
ORD-000008	C-0715	11/11/25	Website	Buy Now Pay	Southeast	3	Sports	71.03	10	7.1	29.3	40.99	0	2.15	Yes	63.93	0	72.44	-72.44
ORD-000009	C-0221	1/25/25	Website	PayPal	Midwest	1	Sports	23.98	15	3.6	20.6	13.76	0	0.89	No	0	20.38	35.25	-14.87
ORD-000010	C-0726	2/3/25	Website	Buy Now Pay	Northeast	1	Food & Bever	345.27	25	86.32	24.66	101.85	0	7.81	No	0	258.95	134.32	124.63
ORD-000011	C-0413	11/29/24	Marketplace	Credit Card	Midwest	1	Books	26.5	5	1.32	5.5	8.06	3.78	1.03	No	0	25.18	18.37	6.81
ORD-000012	C-0759	5/20/24	Social Comm	Credit Card	Northeast	1	Clothing	62.7	0	0	7.5	20.79	5.02	2.12	No	0	62.7	35.43	27.27
ORD-000013	C-0090	6/2/25	Social Comm	PayPal	Midwest	3	Beauty	136.77	10	13.68	31.86	56.88	9.85	3.87	No	0	123.09	102.46	20.63
ORD-000014	C-0632	12/27/25	Mobile App	Credit Card	West Coast	2	Sports	160.31	0	0	15.24	75.89	0	4.95	No	0	160.31	96.08	64.23
ORD-000015	C-0777	6/21/24	Marketplace	Credit Card	Southeast	5	Clothing	254.7	10	25.47	59.06	106.5	34.38	6.95	No	0	229.23	206.89	22.34
ORD-000016	C-0559	6/28/25	Website	PayPal	Midwest	3	Food & Bever	165.86	0	0	28.66	52.32	0	5.11	No	0	165.86	86.09	79.77
ORD-000017	C-0608	1/6/25	Website	Credit Card	Midwest	1	Toys	114.58	0	0	26.36	41.44	0	3.62	No	0	114.58	71.42	43.16
ORD-000018	C-0639	2/23/25	Mobile App	Buy Now Pay	Southwest	2	Beauty	43.21	0	0	23.8	27.23	0	1.55	No	0	43.21	52.58	-9.37
ORD-000019	C-0600	3/16/24	Marketplace	PayPal	Northeast	2	Books	29.95	0	0	33.06	13.54	4.49	1.17	No	0	29.95	52.26	-22.31
ORD-000020	C-0309	7/8/25	Marketplace	PayPal	Northeast	1	Sports	91.38	0	0	23.16	56.42	13.71	2.95	No	0	91.38	96.24	-4.86
ORD-000021	C-0392	1/15/25	Website	Gift Card	Midwest	2	Beauty	56.46	0	0	19.88	19.09	0	1.94	No	0	56.46	40.91	15.55
ORD-000022	C-0208	1/14/25	Social Comm	Credit Card	West Coast	1	Sports	77.62	0	0	22.84	33.1	6.21	2.55	No	0	77.62	64.7	12.92
ORD-000023	C-0162	12/3/25	Website	Credit Card	Southwest	1	Electronics	192.38	0	0	10.06	72.59	0	5.88	Yes	192.38	0	88.53	-88.53
ORD-000024	C-0634	5/31/25	Mobile App	Gift Card	Northeast	3	Beauty	233.9	15	35.09	36.82	87.78	0	6.07	No	0	198.81	130.67	68.14
ORD-000025	C-0334	7/17/24	Mobile App	PayPal	West Coast	2	Sports	129.38	10	12.94	16.9	49.49	0	3.68	No	0	116.44	70.07	46.37
ORD-000026	C-0569	9/12/25	Social Comm	Credit Card	Midwest	2	Home & Kitch	58.12	0	0	23.56	21.32	4.65	1.99	Yes	58.12	0	51.52	-51.52
ORD-000027	C-0095	9/14/25	Website	Credit Card	Southwest	1	Food & Bever	136.04	10	13.6	6.3	39.68	0	3.85	Yes	122.44	0	49.83	-49.83
ORD-000028	C-0085	9/29/25	Website	Credit Card	Southwest	1	Toys	46.8	5	2.34	9.9	19.21	0	1.59	No	0	44.46	30.7	13.76
ORD-000029	C-0436	8/25/25	Marketplace	Credit Card	Midwest	1	Toys	69.12	0	0	9.1	20.88	10.37	2.3	No	0	69.12	42.65	26.47
ORD-000030	C-0053	2/4/25	Marketplace	Credit Card	Southeast	1	Clothing	193.68	5	9.68	14.58	98.01	27.6	5.64	No	0	184	145.83	38.17
ORD-000031	C-0605	2/12/25	Marketplace	Gift Card	Midwest	2	Home & Kitch	56.58	0	0	24.44	20.88	8.49	1.94	No	0	56.58	55.75	0.83
ORD-000032	C-0535	12/26/24	Website	Buy Now Pay	Southwest	1	Home & Kitch	66.44	5	3.32	17.88	38.92	0	2.13	No	0	63.12	58.93	4.19

Figure 3. Dataset orders.csv

Dataset information:

1. The dataset orders.csv consist of 2000 rows
2. With the order_id has unique ID for each row
3. Has relationship/matching column “category”

month	platform	spend	impressions	clicks	conversions	revenue_attributed	cpc	cpa	roas
2024-01	Google Ads	2676.14	303594	13271	500	43063.16	0.2	5.35	16.09
2024-01	Facebook Ads	1971.43	338022	7163	333	15950.86	0.28	5.92	8.09
2024-01	Instagram Ads	2418.95	561361	18729	748	54454.46	0.13	3.23	22.51
2024-01	TikTok Ads	1141.7	554273	6967	346	22369.23	0.16	3.3	19.59
2024-01	Email Marketing	868.82	29509	1215	58	4223.97	0.72	14.98	4.86
2024-01	Influencer	2229.89	324906	18550	672	32663.79	0.12	3.32	14.65
2024-02	Google Ads	6122.11	497660	15128	1065	110409.49	0.4	5.75	18.03
2024-02	Facebook Ads	3681.15	518814	14679	848	84633.22	0.25	4.34	22.99
2024-02	Instagram Ads	1667.58	343834	6693	427	38473.67	0.25	3.91	23.07
2024-02	TikTok Ads	943.7	322885	6717	471	37368.13	0.14	2.0	39.6
2024-02	Email Marketing	860.15	18261	811	63	6865.91	1.06	13.65	7.98
2024-02	Influencer	1902.35	196661	14807	855	91158.26	0.13	2.22	47.92
2024-03	Google Ads	7708.44	1101566	47333	3387	215294.41	0.16	2.28	27.93
2024-03	Facebook Ads	2513.96	413529	6540	254	30308.05	0.38	9.9	12.06
2024-03	Instagram Ads	2118.63	418139	13241	493	55146.65	0.16	4.3	26.03
2024-03	TikTok Ads	2936.85	1113686	19229	1518	128845.37	0.15	1.93	43.87
2024-03	Email Marketing	532.24	25622	1394	75	8860.06	0.38	7.1	16.65
2024-03	Influencer	3044.92	310912	20703	456	42953.97	0.15	6.68	14.11
2024-04	Google Ads	7846.89	639155	27881	831	96956.59	0.28	9.44	12.36
2024-04	Facebook Ads	4664.76	894426	19779	937	39426.57	0.24	4.98	8.45
2024-04	Instagram Ads	2344.92	484252	12115	691	31193.79	0.19	3.39	13.3
2024-04	TikTok Ads	2272.7	555474	8215	412	27601.49	0.28	5.52	12.14

Figure 4. Dataset marketing_spend.csv

Dataset information:

1. The dataset products.csv consist of 144 rows
2. With data range in 2024 – 2025 (12 months)

BREAKDOWN BUSINESS QUESTIONS

DESCRIPTIVE ANALYTICS

1. What is the average profit margin by product category?

```
105 •   SELECT
106         category,
107         ROUND(AVG(profit), 2) AS avg_profit
108     FROM orders o
109     INNER JOIN products p
110         ON o.primary_category = p.category
111     GROUP BY category
112     ORDER BY avg_profit DESC;
```

100% 12:117

Result Grid Filter Rows: Search Export:

category	avg_profit
Electronics	52.34
Toys	34.19
Home & Kitchen	33.44
Food & Beverage	30.74
Sports	26.02
Clothing	21.41
Beauty	15.49
Books	9.41

Electronics category is the highest average profit with $52.34, while Books is the lowest average profit with $9.41

2. How does profitability differ across sales channels?

```
118 •   SELECT
119         `channel`,
120         ROUND(AVG(profit), 2) AS avg_profit
121     FROM orders
122     GROUP BY `channel`
123     ORDER BY avg_profit DESC;
```

100% 8:127

Result Grid Filter Rows: Search Export:

channel	avg_profit
Mobile App	36.32
Website	31.60
Social Commerce	17.11
Marketplace	15.40

Mobile App is the highest profitability with $36.32, while Marketplace is the lowest profitability with $15.40

3. What is the return rate by category and channel?

```sql
SELECT
    primary_category,
    `channel`,
    COUNT(*) AS total_orders,
    SUM(CASE WHEN returned = 'Yes' THEN 1 ELSE 0 END) AS returned_orders,
    ROUND(AVG(CASE WHEN returned = 'Yes' THEN 1.0 ELSE 0.0 END) * 100, 2) AS returned_rate
FROM orders
GROUP BY primary_category, `channel`
ORDER BY returned_rate DESC;
```

100% 1:126

Result Grid Filter Rows: Search Export:

primary_category	channel	total_orders	returned_orders	returned_rate
Clothing	Social Commerce	30	4	13.33
Books	Social Commerce	23	3	13.04
Home & Kitchen	Social Commerce	23	3	13.04
Food & Beverage	Social Commerce	24	3	12.50
Clothing	Marketplace	60	7	11.67
Electronics	Website	101	10	9.90
Sports	Mobile App	95	9	9.47

Social Commerce channel is the top 4 high return rate followed by Categories: Clothing, Books, Home & Kitchen and Food & Beverage with returned rate 13.33, 13.04, 13.04 and 12.50, while Marketplace channel is the top 2 low return rate followed by Categories: Food & Beverage and Sports with returned rate 1.61 and 1.92

4. Estimate how much total revenue was lost to returns over the analysis period !

```sql
SELECT
    SUM(refund_amount) AS total_revenue_lost_to_return
FROM orders
WHERE returned = 'Yes';
```

100% 1:152

Result Grid Filter Rows: Search Export:

total_revenue_lost_to_return
20582.45

Total revenue was lost to returns over the analysis period is $20,582

5. Which advertising platform delivering the best ROAS?

```sql
SELECT
    platform,
    SUM(spend) AS total_spend,
    SUM(revenue_attributed) AS total_revenue,
    ROUND(SUM(revenue_attributed) / SUM(spend), 2) AS overall_roas
FROM marketing_spend
GROUP BY platform
ORDER BY overall_roas DESC;
```

100% 58:153

Result Grid Filter Rows: Search Export:

platform	total_spend	total_revenue	overall_roas
TikTok Ads	57229.22	1374627.17	24.02
Influencer	97663.12	2216974.26	22.70
Instagram Ads	65154.02	1024639.06	15.73
Google Ads	152546.48	2194120.83	14.38
Facebook Ads	106451.93	1218572.43	11.45
Email Marketing	24461.37	117681.45	4.81

Tiktok Ads is delivering the best ROAS with 24.02%, while Email Marketing is delivering the worst ROAS with 4.81%

DIAGNOSTICS ANALYTICS

1. Which categories are the most and least profitable, and what is driving the difference (product cost, shipping, returns, or discounts)?

```
169    # -- Business Contribution
170 •  SELECT
171        primary_category,
172        COUNT(order_id) AS total_order,
173        SUM(profit) AS total_profit
174    FROM orders
175    GROUP BY primary_category
176    ORDER BY total_profit DESC;
177
```

100% 44:167

Result Grid Filter Rows: Search Exp

primary_category	total_order	total_profit
Electronics	267	13973.46
Toys	257	8786.25
Sports	292	7596.63
Food & Beverage	247	7591.77
Home & Kitchen	200	6688.57
Clothing	293	6272.48
Beauty	205	3174.88
Books	239	2249.79

Electronics total order = 267 and total profit = $ 13,973 , while Books total order = 239 and total profit = $ 2,250

```
180    # -- Order-Level Profitability
181 •  SELECT
182        primary_category,
183        ROUND(AVG(profit), 2) AS avg_profit_per_order,
184        ROUND(SUM(profit) / SUM(net_revenue) * 100, 2) AS profit_margin
185    FROM orders
186    GROUP BY primary_category
187    ORDER BY profit_margin DESC;
188
```

100% 1:177

Result Grid Filter Rows: Search Export:

primary_category	avg_profit_per_order	profit_margin
Electronics	52.34	31.13
Toys	34.19	26.15
Home & Kitchen	33.44	25.37
Food & Beverage	30.74	24.76
Sports	26.02	23.50
Clothing	21.41	19.99
Beauty	15.49	17.39
Books	9.41	11.94

Electronics profit per order = $ 52.34, while Books profit per order = $ 9.41. Electronics is the most profitable category, with a 31.13% profit margin, while Books is the least profitable at 11.94%.

```sql
# -- Unit-Level Profitability
SELECT
    primary_category,
    SUM(items_ordered) AS total_units,
    ROUND(SUM(gross_revenue) / SUM(items_ordered), 2) AS revenue_per_unit,
    ROUND(SUM(profit) / SUM(items_ordered), 2) AS profit_per_unit
FROM orders
GROUP BY primary_category
ORDER BY profit_per_unit DESC;
```

100% 1:188

Result Grid Filter Rows: Search Export:

primary_category	total_units	revenue_per_unit	profit_per_unit
Electronics	507	104.29	27.56
Home & Kitchen	382	80.84	17.51
Toys	503	77.70	17.47
Food & Beverage	446	82.69	17.02
Sports	528	70.93	14.39
Clothing	554	67.72	11.32
Beauty	382	54.64	8.31
Books	456	49.15	4.93

Electronics revenue per unit = $ 104.29, while Books revenue per unit = $ 49.15. Electronics profit per unit = $ 27.56, while Books profit per unit = $ 4.93

```sql
## Driving the difference (product cost, shipping, returns, or discounts)
SELECT
    primary_category,
    ROUND(AVG(net_revenue), 2) AS avg_net_revenue,
    ROUND(AVG(product_cost), 2) AS avg_product_cost,
    ROUND(SUM(product_cost) / SUM(gross_revenue) * 100, 2) AS product_cost_pct,
    ROUND(AVG(shipping_cost), 2) AS avg_shipping_cost,
    ROUND(SUM(shipping_cost) / SUM(gross_revenue) * 100, 2) AS shipping_cost_pct,
    ROUND(AVG(refund_amount), 2) AS avg_refund_amount,
    ROUND(SUM(refund_amount) / SUM(gross_revenue) * 100, 2) AS refund_pct,
    ROUND(AVG(discount_amount), 2) AS avg_discount_amount,
    ROUND(SUM(discount_amount) / SUM(gross_revenue) * 100, 2) AS discount_pct
FROM orders
WHERE primary_category IN ('Electronics', 'Books')
GROUP BY primary_category;
```

100% 79:203

Result Grid Filter Rows: Search Export:

primary_category	avg_net_revenue	avg_product_cost	product_cost_pct	avg_shipping_cost	shipping_cost_pct	avg_refund_amount	refund_pct	avg_discount_amount	discount_pct
Electronics	168.11	75.25	38.00	26.74	13.50	15.27	7.71	14.65	7.40
Books	78.86	37.48	39.96	26.20	27.94	8.81	9.40	6.11	6.51

Electronics is **the most profitable category**, while Books is **the least profitable**. The difference is not primarily driven by order volume, as both categories have a similar number of orders. Books has a substantially higher shipping-cost burden relative to revenue (27.94% vs. 13.50%), followed by a higher refund burden (9.40% vs. 7.71%) and slightly higher product-cost burden (39.96% vs. 38.00%). Meanwhile, Electronics has a slightly higher discount burden but still achieves much higher profitability. Therefore, shipping costs appear to be the

strongest contributing factor associated with the profitability gap, with refunds and product costs as secondary factors.

2. Which channel has the best and worst profit per order after accounting for platform fees?

```
237 •   SELECT
238         `channel`,
239         COUNT(order_id) AS total_orders,
240         ROUND(AVG(items_ordered), 2) AS avg_items_ordered,
241         ROUND(AVG(profit), 2) AS avg_profit_per_order,
242         ROUND(AVG(platform_fee), 2) AS avg_platform_fee
243     FROM orders
244     GROUP BY `channel`
245     ORDER BY avg_profit_per_order DESC;
```

100% 13:233

Result Grid Filter Rows: Search Export:

channel	total_orders	avg_items_ordered	avg_profit_per_order	avg_platform_fee
Mobile App	589	1.92	36.32	0.00
Website	795	1.85	31.60	0.00
Social Commerce	197	1.91	17.11	9.87
Marketplace	419	1.87	15.40	18.97

Mobile App has the best profit per order with avg_profit_per_order $36.32, while Marketplace has the worst profit per order with avg_profit_per_order $15.40 after accounting for platform fees. Marketplace has the higher platform fee than others this cause it's the worst profit per order after accounting for platform fees.

3. Are there any platforms where the company is spending money but not getting a positive return?

```
253 •   SELECT
254         platform,
255         ROUND(SUM(spend), 2) AS total_spend,
256         ROUND(SUM(revenue_attributed), 2) AS total_revenue,
257         ROUND(SUM(revenue_attributed) / SUM(spend), 2) AS overall_roas,
258         ROUND(SUM(revenue_attributed) - SUM(spend), 2) AS revenue_minus_spend
259     FROM marketing_spend
260     GROUP BY platform
261     ORDER BY overall_roas;
```

100% 1:267

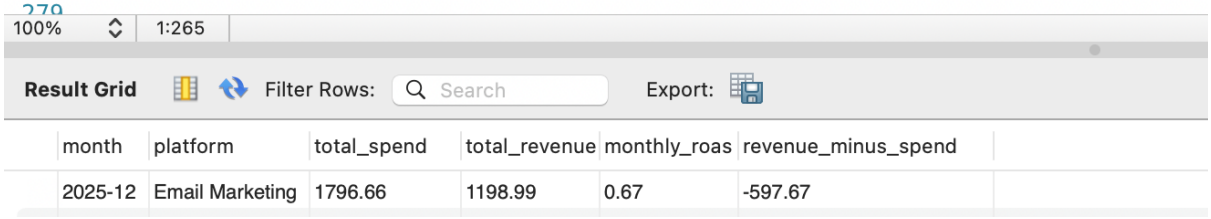
Result Grid Filter Rows: Search Export:

platform	total_spend	total_revenue	overall_roas	revenue_minus_spend
Email Marketing	24461.37	117681.45	4.81	93220.08
Facebook Ads	106451.93	1218572.43	11.45	1112120.50
Google Ads	152546.48	2194120.83	14.38	2041574.35
Instagram Ads	65154.02	1024639.06	15.73	959485.04
Influencer	97663.12	2216974.26	22.70	2119311.14
TikTok Ads	57229.22	1374627.17	24.02	1317397.95

Overall, all six marketing platforms generated a positive ROAS over the analysis period with Email Marketing having the lowest ROAS at 4.81 and TikTok Ads the highest at 24.02.

Let's take a look for a monthly roas

```sql
SELECT
    `month`,
    platform,
    ROUND(SUM(spend), 2) AS total_spend,
    ROUND(SUM(revenue_attributed), 2) AS total_revenue,
    ROUND(SUM(revenue_attributed) / SUM(spend), 2) AS monthly_roas,
    ROUND(SUM(revenue_attributed) - SUM(spend), 2) AS revenue_minus_spend
FROM marketing_spend
GROUP BY `month`, platform
HAVING SUM(revenue_attributed) < SUM(spend)
ORDER BY monthly_roas;
```



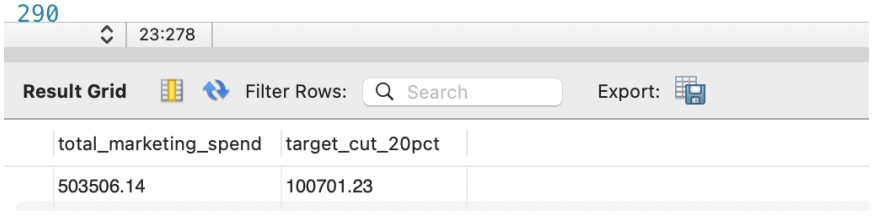
month	platform	total_spend	total_revenue	monthly_roas	revenue_minus_spend
2025-12	Email Marketing	1796.66	1198.99	0.67	-597.67

However, monthly analysis shows that Email Marketing had a negative return in December 2025, with a ROAS of 0.67.

PRESCRIPTIVE ANALYTICS

1. If the CEO asked you to cut 20% of the marketing budget, which platforms and months would you recommend reducing spend on? Support your recommendation with data.

```sql
SELECT
    ROUND(SUM(spend), 2) AS total_marketing_spend,
    ROUND(SUM(spend) * 0.20, 2) AS target_cut_20pct
FROM marketing_spend;
```



total_marketing_spend	target_cut_20pct
503506.14	100701.23

Total marketing spend over the analysis period is $503,506, while target cut 20% of marketing budget is $100,701

```sql
WITH ranked AS (
    SELECT
        `month`,
        platform,
        SUM(spend) AS total_spend,
        SUM(revenue_attributed) AS total_revenue,
        SUM(revenue_attributed) / SUM(spend) AS monthly_roas,
        SUM(SUM(spend)) OVER () * 0.20 AS target_cut,
        COALESCE(
            SUM(SUM(spend)) OVER (
                ORDER BY
                    SUM(revenue_attributed) / SUM(spend) ASC,`month`,platform
                ROWS BETWEEN UNBOUNDED PRECEDING AND 1 PRECEDING
            ),0
        ) AS previous_cumulative_spend
    FROM marketing_spend
    GROUP BY `month`, platform
)
SELECT
    `month`,
    platform,
    ROUND(total_spend, 2) AS total_spend,
    ROUND(total_revenue, 2) AS total_revenue,
    ROUND(monthly_roas, 2) AS monthly_roas,
    ROUND(LEAST(total_spend,GREATEST(0, target_cut - previous_cumulative_spend)), 2) AS recommended_cut
FROM ranked
WHERE previous_cumulative_spend < target_cut
ORDER BY monthly_roas ASC;
```

0% 1:302

Result Grid Filter Rows: Search Export:

month	platform	total_spend	total_revenue	monthly_roas	recommended_cut
2025-07	Email Marketing	1362.00	2364.53	1.74	1362.00
2024-06	Email Marketing	942.79	1694.39	1.80	942.79
2025-09	Facebook Ads	3877.76	9733.42	2.51	3877.76
2024-10	Email Marketing	1040.57	2620.57	2.52	1040.57
2024-08	Facebook Ads	5418.73	16769.26	3.09	5418.73
2025-06	Email Marketing	1293.73	4167.46	3.22	1293.73

To reduce the marketing budget by 20%, the company should prioritize reducing spending on Email Marketing in December 2025, October 2025, July 2025, and June 2024, followed by selected Facebook Ads campaigns with relatively low monthly ROAS.

DEEPER ANALYSIS USING PYTHON

1. Descriptive Statistics

```
profit_summary = orders["profit"].describe().to_frame("profit")

display(profit_summary.round(2))

print("Skewness:", round(orders["profit"].skew(), 3))
```

	profit
count	2000.00
mean	28.17
std	69.55
min	-469.09
25%	-2.36
50%	15.18
75%	50.96
max	465.35

```
Skewness: 0.572
```

The dataset contains 2,000 observations, with an average profit of 28.17 and a median of 15.18. The standard deviation of 69.55 indicates substantial variability in profit. Profit ranges from -469.09 to 465.35, showing that some orders generate losses while others achieve high profits.

The first quartile (Q1) is -2.36, meaning approximately 25% of orders have profit at or below this value, while the third quartile (Q3) is 50.96. The skewness value of 0.572 indicates a moderately right-skewed distribution, suggesting that some high-profit orders pull the mean above the median.

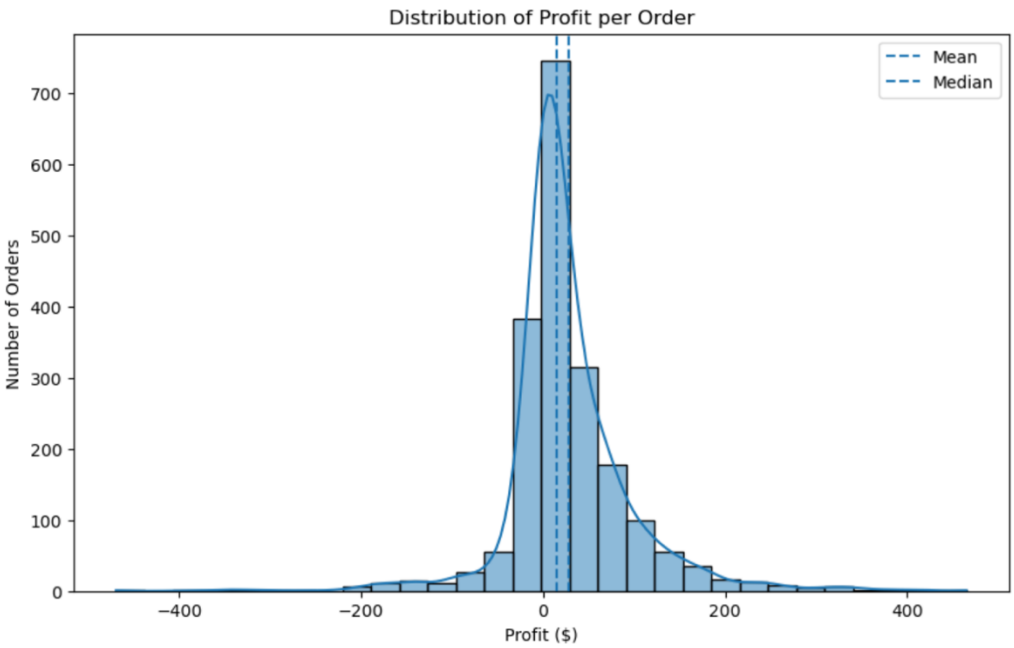


Figure 5. Profit Distribution Matplotlib Seaborn

2. Correlation Analysis

```
corr_cols = [
    "profit",
    "gross_revenue",
    "product_cost",
    "shipping_cost",
    "refund_amount",
    "discount_amount"
]

corr = orders[corr_cols].corr()

display(corr.round(4))
```

	profit	gross_revenue	product_cost	shipping_cost	refund_amount	discount_a
profit	1.0000	0.5711	0.4871	0.1581	-0.5455	
gross_revenue	0.5711	1.0000	0.9647	0.6290	0.2501	
product_cost	0.4871	0.9647	1.0000	0.6938	0.2348	
shipping_cost	0.1581	0.6290	0.6938	1.0000	0.1340	
refund_amount	-0.5455	0.2501	0.2348	0.1340	1.0000	
discount_amount	0.1682	0.5344	0.5162	0.3540	0.0814	

Gross revenue and product cost show moderate positive correlations with profit (r = 0.5711 and r = 0.4871, respectively), while refund amount has a moderate negative correlation with profit (r = -0.5455). Shipping cost and discount amount show only weak correlations with profit (r = 0.1581 and r = 0.1682). This suggests that refund amount may be a more relevant profitability concern than absolute shipping or discount costs, although these correlations do not establish causality.

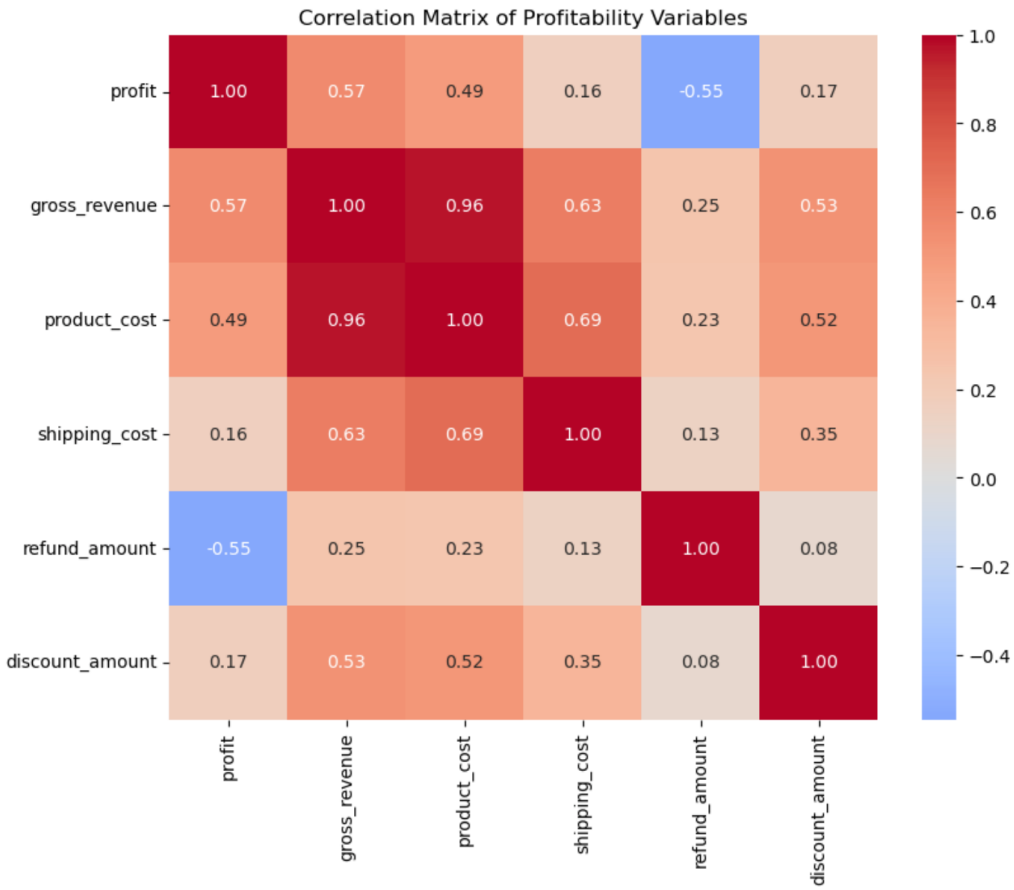


Figure 6. Correlation Matrix Matplotlib Seaborn

3. Loss-making Orders

```
orders[orders["profit"] < 0].groupby("channel")["profit"].agg(
    ["count", "mean", "sum"]
)
```

channel	count	mean	sum
Marketplace	155	-28.126645	-4359.63
Mobile App	145	-30.782759	-4463.50
Social Commerce	63	-32.476032	-2045.99
Website	200	-35.214250	-7042.85

Although Mobile App generates the highest average profit per order, 145 orders were still loss-making, resulting in a total loss of $4,463.50. Website had the largest number of loss-making orders (200) and the highest total loss ($7,042.85), suggesting that profitability varies substantially within each sales channel.

4. Loss-rate by Channel

```
result = orders.groupby("channel")["profit"].agg(
    total_orders="count",
    loss_orders=lambda x: (x < 0).sum()
)

result["loss_rate"] = (
    result["loss_orders"] / result["total_orders"] * 100
).round(2)

result
```

channel	total_orders	loss_orders	loss_rate
Marketplace	419	155	36.99
Mobile App	589	145	24.62
Social Commerce	197	63	31.98
Website	795	200	25.16

Although Mobile App generates the highest average profit per order, 145 orders were still loss-making, resulting in a total loss of $4,463.50. Website had the largest number of loss-making orders (200) and the highest total loss ($7,042.85), suggesting that profitability varies substantially within each sales channel.

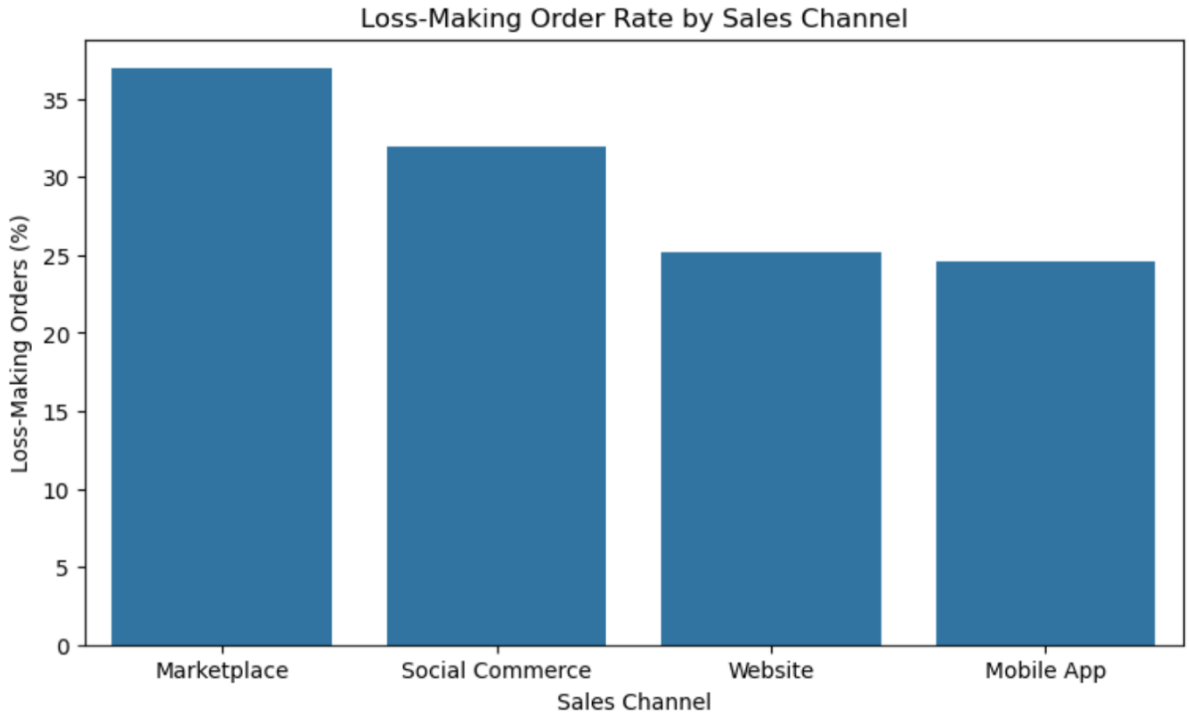


Figure 7. Loss-order Rate by Channel Matplotlib Seaborn

5. Category x Channel Profitability

```
category_channel.sort_values(
    "avg_profit",
    ascending=False
)
```

primary_category	channel	total_orders	avg_profit	total_profit
Electronics	Mobile App	78	74.74	5829.82
Home & Kitchen	Mobile App	63	48.78	3073.26
Electronics	Website	101	48.05	4853.24
Food & Beverage	Website	87	44.82	3899.30
Electronics	Marketplace	65	42.41	2756.94
Toys	Social Commerce	23	38.65	888.89
	Website	111	37.25	4135.21
Beauty	Social Commerce	27	12.41	335.08
Clothing	Social Commerce	30	11.42	342.53
Books	Website	88	11.37	1000.77
Beauty	Marketplace	36	6.23	224.42
Clothing	Marketplace	60	1.17	70.03
Books	Marketplace	51	-3.62	-184.73
	Social Commerce	23	-8.52	-196.03

The highest primary category based on avg_profit is 'Electronics' with Mobile App as channel with $74.74, while the lowest primary category is 'Books' with Social Commerce as channel with $-8.52

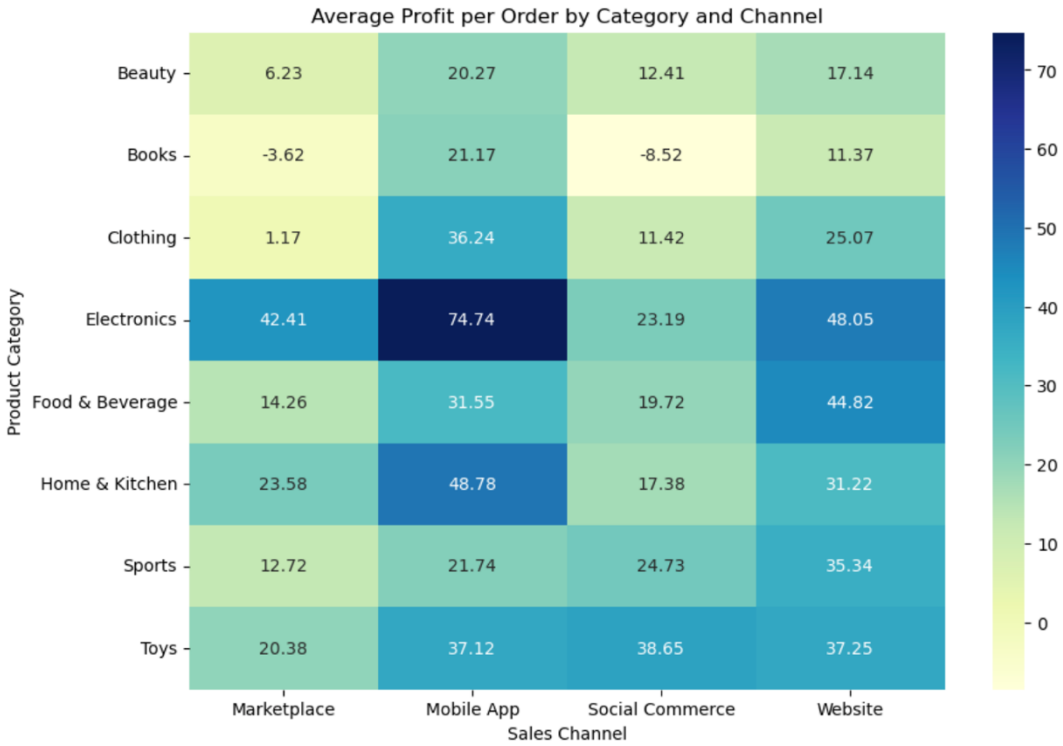


Figure 8. Matrix Avg profit per Order by Category and Channel Matplotlib Seaborn

```
category_channel.sort_values(
    "total_profit",
    ascending=False
)
```

primary_category	channel	total_orders	avg_profit	total_profit
Electronics	**Mobile App**	78	74.74	5829.82
	Website	101	48.05	4853.24
Sports	**Website**	121	35.34	4276.33
Toys	**Website**	111	37.25	4135.21
Food & Beverage	**Website**	87	44.82	3899.30
Clothing	**Website**	134	25.07	3359.39
Clothing	**Social Commerce**	30	11.42	342.53
Beauty	**Social Commerce**	27	12.41	335.08
	Marketplace	36	6.23	224.42
Clothing	**Marketplace**	60	1.17	70.03
Books	**Marketplace**	51	-3.62	-184.73
	Social Commerce	23	-8.52	-196.03

Electronics sold through Mobile App is the most profitable category-channel combination, generating $74.74 average profit per order and $5,829.82 total profit. In contrast, Books show substantial channel-level differences: Mobile App generates $21.17 average profit per order, while Marketplace and Social Commerce generate negative average profits of $-3.62 and $-8.52, respectively.

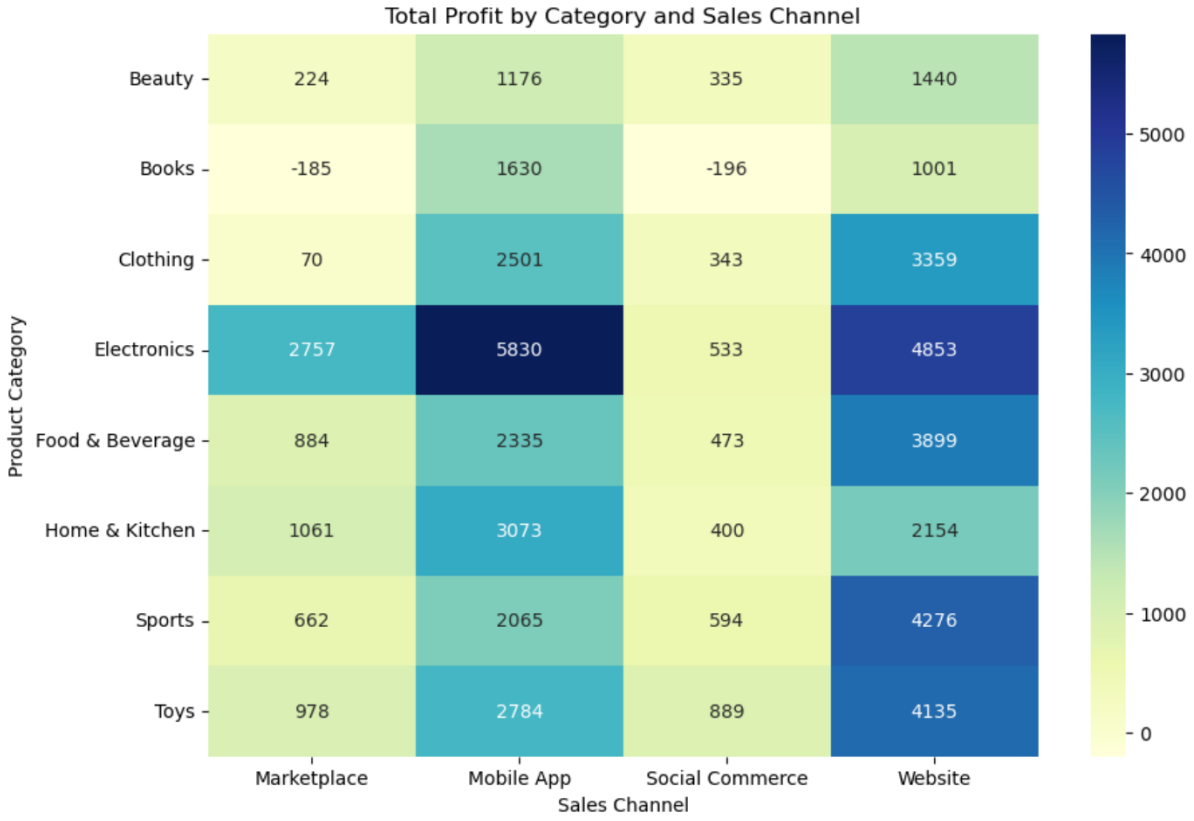


Figure 9. Matrix Total Profit by Category and Channel Matplotlib Seaborn

VISUALIZATION

Using Microsoft Power BI with integrated from MySQL server to build visualization dashboard and storytelling after analyzing the datas.

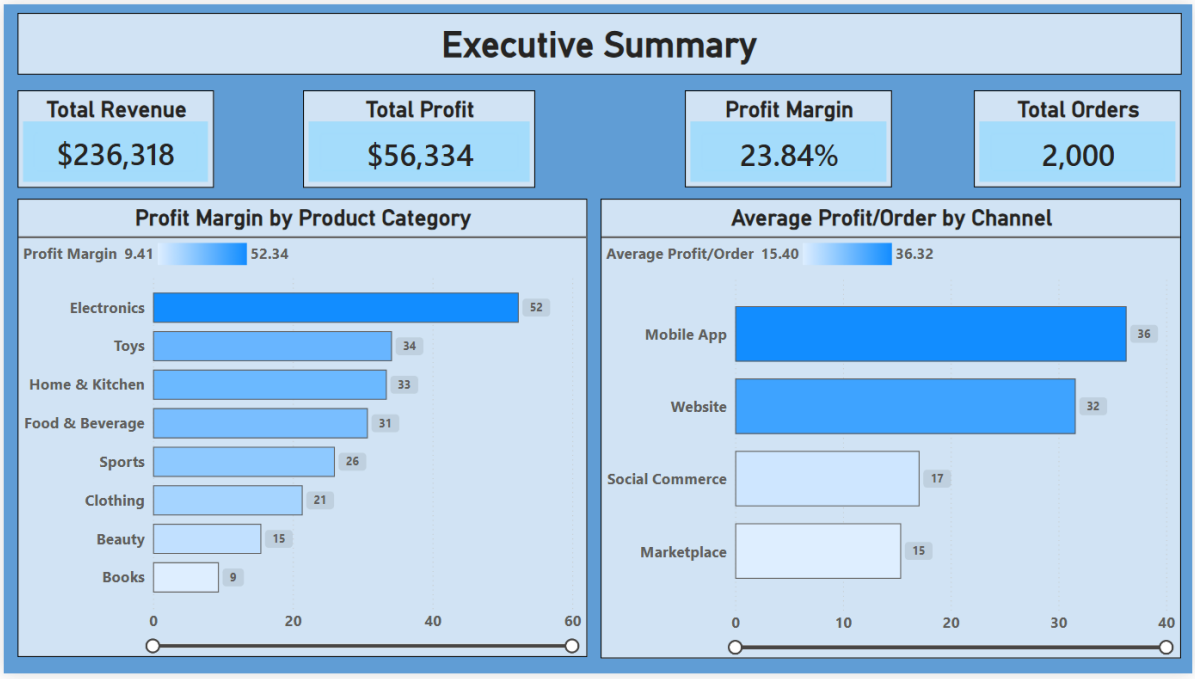


Figure 10. Executive Summary Dashboard Microsoft Power BI

The Executive Summary shows $236,318 in total revenue, $56,334 in total profit, a 23.84% profit margin, and 2,000 total orders. Electronics has the highest profit margin at 52%, while Books has the lowest at 9%. By sales channel, Mobile App generates the highest average profit per order at $36.32, while Marketplace has the lowest at $15.40.

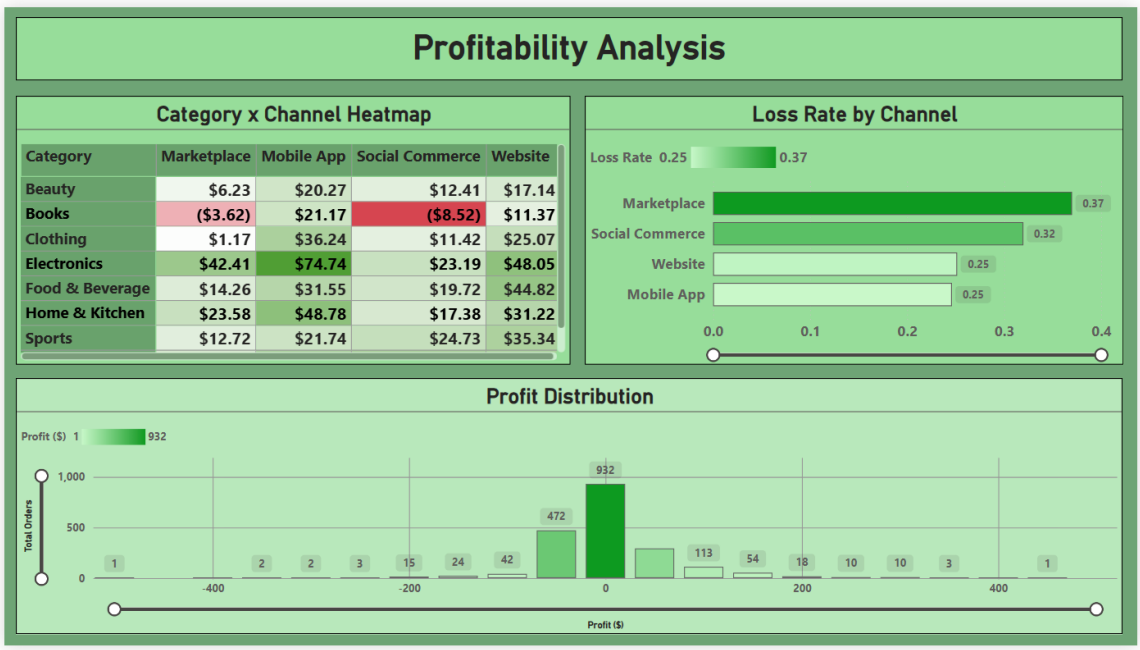


Figure 11. Profitability Analysis Dashboard Microsoft Power BI

Profitability varies considerably across category and channel combinations. Electronics generates the highest average profit per order, particularly through the Mobile App at $74.74, while Books performs poorly on Marketplace and Social Commerce, with average losses of $3.62 and $8.52 per order.

Marketplace has the highest loss rate at 36.99%, followed by Social Commerce at 31.98%. Mobile App has the lowest loss rate at 24.62%, indicating relatively stronger profitability performance across orders.

Profit distribution is concentrated around the break-even range, with 932 orders in the central profit bucket. However, the distribution also includes both significant losses and high-profit orders, indicating substantial variation in profitability across individual orders.

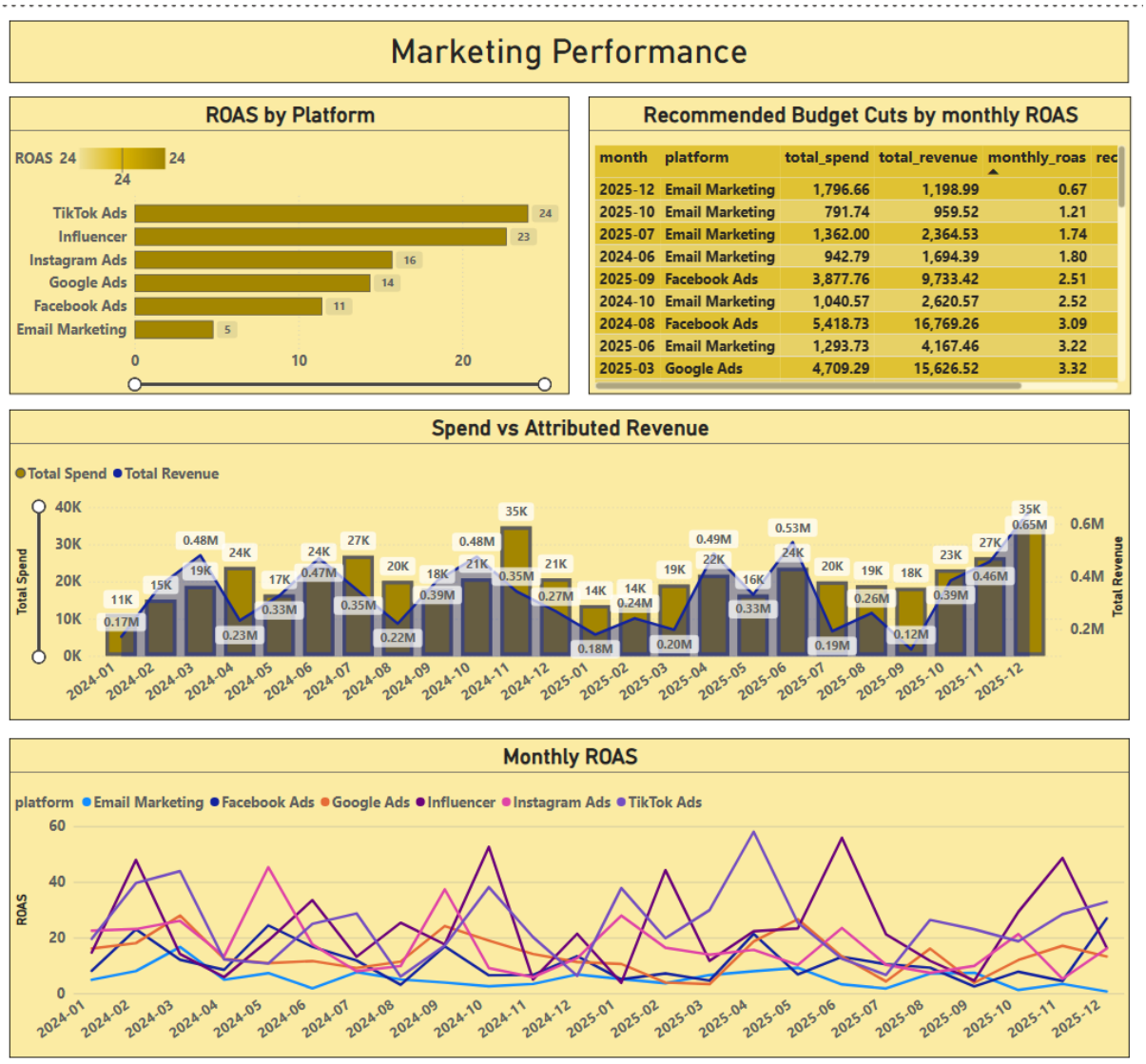


Figure 12. Marketing Performance Dashboard Microsoft Power BI

The Marketing Performance Dashboard shows TikTok Ads has the highest overall ROAS at approximately 24x, followed by Influencer Marketing at 23x and Instagram Ads at 16x. Email Marketing has the lowest overall ROAS at approximately 5x, indicating relatively lower marketing efficiency compared with the other platforms.

Marketing performance varies substantially across months and platforms. Several platforms show sharp fluctuations in monthly ROAS, suggesting that marketing efficiency is not consistent over time and should be evaluated at the month-platform level rather than based only on overall platform performance.

The recommended budget cuts prioritize the lowest-ROAS month-platform combinations. The lowest-performing combination is Email Marketing in December 2025, with a monthly ROAS of 0.67x, followed by Email Marketing in October 2025 at 1.21x and July 2025 at 1.74x. These combinations are prioritized to support the targeted 20% reduction in marketing spend.

Spend and attributed revenue also fluctuate across months, indicating that higher marketing spend does not consistently translate into proportionally higher attributed revenue. This highlights the importance of evaluating marketing efficiency through ROAS rather than spend or revenue alone.

INSIGHTS

BrightCart generated $236,318 in revenue and $56,334 in net profit across 2,000 orders, but profitability varies substantially across categories and channels. Electronics is the strongest-performing category, generating $13,973 in total profit and a 31.13% profit margin, while Books generates only $2,250 in total profit and an 11.94% margin.

The profitability gap between Electronics and Books is primarily associated with cost structure rather than order volume. Both categories have a similar number of orders, but Books has a significantly higher shipping-cost burden at 27.94% of gross revenue compared with 13.50% for Electronics. Books also has higher refund and product-cost burdens, at 9.40% and 39.96%, respectively, compared with 7.71% and 38.00% for Electronics. This indicates that shipping costs are the strongest contributor to the observed profitability gap.

Sales channel performance also differs significantly. Mobile App generates the highest average profit per order at $36.32, while Marketplace generates the lowest at $15.40 after accounting for platform fees. At the category-channel level, Electronics on Mobile App is particularly strong at $74.74 average profit per order, while Books generates losses on Marketplace and Social Commerce. This shows that profitability depends not only on the product category or channel individually, but also on their combination.

Returns and individual-order profitability are important sources of margin pressure. The analysis estimates $20,582 in revenue lost through refunds, with Social Commerce and several categories showing relatively high return rates. Python analysis also shows substantial variation in order-level profit, with an average profit of $28.17 compared with a median of $15.19 and a range from -$469.09 to $465.35. Refund amount has a moderate negative correlation with profit, suggesting that higher refund amounts are associated with lower profitability.

Marketing performance is positive overall, but efficiency varies considerably. TikTok Ads has the highest overall ROAS at 24.02x, while Email Marketing has the lowest at 4.81x. Although all platforms are positive on an aggregate basis, monthly analysis reveals underperforming periods, including Email Marketing in December 2025 with a ROAS of 0.67x.

RECOMMENDATIONS

Prioritize high-profit category-channel combinations, particularly Electronics through the Mobile App, while reviewing low-performing combinations such as Books on Marketplace and Social Commerce. For weaker combinations, management should investigate pricing, shipping costs, product costs, discounts, and returns before deciding whether to reduce their exposure.

Focus on reducing the shipping-cost burden for lower-margin categories, especially Books. Since shipping represents 27.94% of gross revenue for Books versus 13.50% for Electronics, potential actions include optimizing fulfillment, negotiating shipping rates, consolidating shipments, or reviewing minimum-order thresholds for free shipping. e-commerce_analysis

Reduce return-related margin leakage by investigating the categories and channels with higher return rates, particularly Social Commerce and categories such as Clothing, Books, Home & Kitchen, and Food & Beverage. Rather than simply reducing sales volume, the company should identify the underlying reasons for returns and target operational or product-level improvements. The $20,582 in refund-related revenue loss makes returns a meaningful profitability lever. e-commerce_analysis

Implement a selective 20% marketing budget reduction rather than cutting every platform equally. The total marketing spend is $503,506, giving a target reduction of approximately $100,701. The analysis recommends starting with the lowest-ROAS month-platform combinations, particularly Email Marketing in December 2025, October 2025, July 2025, and June 2024, followed by selected Facebook Ads campaigns with relatively low monthly ROAS. e-commerce_analysis

Protect and continue monitoring high-performing marketing channels such as TikTok Ads, while evaluating Email Marketing at a more granular monthly level. Platform-level ROAS alone can hide periods of poor performance, so future budget allocation should be based on month-platform performance rather than overall ROAS alone.